Local research workbench · UI MVP

Run the workflow. Check the evidence. Keep the limits visible.

A practical manual for setup, the first synthetic run, result review, and recovery. Follow one action at a time. Check the expected result before you continue.

First successful flight

Install the locked environment. Start the local server. Create a synthetic study. Run `heuristic-hu`. Open the HTML research report.

No patient data, model download, or AI server is needed for this first run.

> **Research use only.** This is a self-made experimental tool created with AI assistance, to the best of our knowledge. It is not medical advice from a qualified healthcare professional, a diagnosis, or a medical device. Findings may be incorrect, incomplete or misleading. Do not use the output for diagnosis or treatment decisions.

Choose your checklist

1. Preflight: install and start
2. The browser workflow: input to offline report
3. Read findings, severity and review context
4. Repeat the workflow from the command line
5. Configure paths and local models
6. Protect data, recover and shut down

Author: Marcel Petrick · GPL-3.0-or-later · Project generated with AI assistance.
Written in a flight-manual style with short actions and explicit checkpoints; no formal technical-English compliance claim. This manual describes the current MVP, not the planned benchmark or LAN runner.

Install. Verify. Start locally.

Use a terminal on the computer that will run the software. The commands below assume a source checkout and `uv`. Replace the example path with your checkout path. If `uv` is missing, follow the official uv installation instructions first.

1. **Open the repository directory.**

   ```
   cd /home/mpetrick/repos/ThoraxGPT_from_Wish
   ```

2. **Install the locked Python environment.**

   ```
   uv sync --frozen --all-groups
   ```

   Python 3.14 or newer is required. `uv` can provision the requested Python environment. Initial installation needs package access; inference stays local.

3. **Check the installed command.**

   ```
   uv run thoraxbench --version
   uv run thoraxbench --help
   ```

   Expect a version, the research disclaimer, and the command list.

4. **Start the workbench.**

   ```
   uv run thoraxbench serve
   ```

   Leave this terminal open. Open http://127.0.0.1:8000 on the same computer.

> **Checkpoint:** The Analysis page appears in English and light mode. The page shows the research disclaimer. A new installation may have no studies yet. This is expected.

Required for the demo

A supported Python environment, the locked dependencies, a browser, and enough free disk space for synthetic data and outputs.

Optional for vision models

A local Ollama installation and compatible model weights. The deterministic `heuristic-hu` baseline works without them.

Port 8000 is already in use

```
uv run thoraxbench serve --port 8001
```

Open `http://127.0.0.1:8001`. The server binds to loopback. LAN access and remote inference are not supported in this MVP.

Complete your first browser run.

Create demo → **Inspect study** → **Run locally** → **Review** → **Export**

1. **Select “Create synthetic demo”.**

   Expect a synthetic study with 64 axial slices. It contains simulated nodule and fluid-density regions. The images contain no patient data.

2. **Inspect the study viewer.**

   Move the slice slider. Use the Left and Right arrow keys when the slider has focus. Change the display window to Lung, Mediastinum, Bone or Abdomen. Window changes affect the viewer only.

3. **Select “heuristic-hu” as the model.**

   This baseline uses HU-volume measurements. It sends no rendered images to a vision model. The image-count control does not affect its predictions.

4. **Select “Run analysis locally”.**

   Wait for a completed result or explicit failure/abstention. Expect synthetic nodule and effusion findings on the default demo. Exact timings vary by hardware.

5. **Read the status before you read the findings.**

   Check study quality, the finding table, severity/review context and the English research report. Follow the interpretation checklist on the next page.

6. **Select “Download HTML”.**

   Open the saved file in a browser. Check the findings and embedded image gallery. Also save JSON for structured output and “Download report” for Markdown.

> **Checkpoint:** The HTML report opens after you stop the server. It needs no external images, fonts, scripts or internet access. Missing evidence references remain explicit.

Keyboard help and preferences

Use Tab to reach controls and help-bearing fields. Focus or hover shows translated help; Escape dismisses it. The language switch selects English or German. The theme switch selects light or dark. Both choices persist in this browser. Research reports and model prompts remain English.

About

Open About to inspect the author, application version, research purpose and installed runtime libraries with their declared licenses.

> **Privacy checkpoint:** A downloaded report contains images and research identifiers. Keep real-study exports in controlled local storage. The browser may save downloads outside the repository; manage those copies as sensitive data.

Inspect claims before conclusions.

Review every model claim against its evidence. The software cannot decide that a person is healthy, that care is unnecessary, or that a finding is real.

Read first	What it means	Operator action
Completed	The engine returned output. This does not validate it.	Inspect QC, findings, evidence and provenance.
Abstained / failed	No usable research reading is available.	Read the reason. Preserve the failed run. Do not interpret it as “nothing found”.
Present / not detected / uncertain	The model's reported presence status.	Treat negative and uncertain output as fallible.
Presence score	Uncalibrated support for presence; not measured accuracy.	Do not treat 0.79 as a validated 79% probability.
Severity / typical attention	Educational reference context, not validated triage.	Check its assumptions and sources. Do not derive personal medical instructions.

Severity and “should this be checked?”

The reference scale runs from **incidental** → **low** → **moderate** → **high** → **critical**. Typical attention uses **routine, soon, urgent, emergency**. These describe general reference knowledge for a reported label. They do not establish the severity or urgency of an actual person's condition.

> **No automatic clearance:** An absent, low-rated, unknown or uncertain finding does not establish that no professional review is needed. The tool cannot answer whether a person should be checked. Its output is not advice from a healthcare practitioner.

Check image evidence

- Vision-model reports show the actual prepared input images. **Image 1** is the first model image. A sampled image set may miss other parts of the study.
- Heuristic reports show **illustrative source-slice previews**. Their source indices start at **0**; they are not vision-model inputs.
- An evidence link is a reported association, not verified localization or a segmentation. Invalid or missing references must remain unresolved.

Check reproducibility

Save the JSON and read the original response, QC, timings, prompt/configuration hashes and provenance. Analysis-stage timing excludes DICOM loading and reporting. A working synthetic run or local-model transport test is not a real-patient accuracy benchmark.

Repeat a run and save its artifacts.

Run these commands from the repository root. Keep the working directory unchanged between steps. These examples use only synthetic data.

1. **Create a fresh synthetic study.**

```
uv run thoraxbench demo-data data/demo/manual
```

2. **Run the deterministic baseline.**

```
uv run thoraxbench analyze data/demo/manual \
  --model heuristic-hu \
  --output data/results/manual/result.json \
  --report data/results/manual/report.md \
  --html data/results/manual/report.html
```

3. **Open the saved HTML file.**

   Open `data/results/manual/report.html` in your browser. Inspect the matching JSON and Markdown files in the same folder.

> **Checkpoint:** A successful run returns exit status 0. A failed run or abstention returns a nonzero status. When a result exists, saved artifacts explain the abstention. Failures before result creation may produce only an error message.

Optional: run a local vision model

First check availability. A configured model is not necessarily installed, licensed for your intended use, or accurate.

```
uv run thoraxbench models
uv run thoraxbench analyze data/demo/manual \
  --model medgemma1.5-q4 --n 4 \
  --output data/results/manual/vision.json \
  --report data/results/manual/vision.md \
  --html data/results/manual/vision.html
```

The model must already exist in local Ollama. Larger image requests may take minutes. Empty predictions, truncated output, invalid responses and unsuitable inputs can cause abstention.

Available overrides

`--model` selects an ID from `configs/models/`. `--prompt` selects P0–P4. `--strategy` overrides the sampling strategy. `--n` requests a positive number of sampled slices; the browser limits it to 1–16. Use `uv run thoraxbench analyze --help` for the current options.

HTML output must be inside the configured data directory's `results/` folder. Destinations or symlinks that escape that folder are rejected before inference. JSON and Markdown examples also stay inside ignored results storage.

Keep paths explicit and inference local.

Set environment variables before starting the command or server. Existing data and manifests do not move when you change a setting.

Environment variable	Default / role
`THORAXBENCH_DATA_DIR`	`<checkout>/data`. Raw data, caches, splits, sealed IDs, demos and results. Keep medical data outside Git.
`THORAXBENCH_OLLAMA_URL`	`http://localhost:11434`. Only loopback inference endpoints are accepted.
`THORAXBENCH_DATA_BUDGET_GB`	`10.0` decimal GB. Configured data budget; inspect actual usage before writing. Current analysis/demo routes do not provide a complete quota enforcement system.
`THORAXBENCH_CT_CLIP_URL` `THORAXBENCH_M3D_URL`	`http://localhost:8011` / `:8012`. Optional CT-CLIP / M3D services.

`THORAXBENCH_CT_CHAT_URL`
`THORAXBENCH_RADFM_URL`
`THORAXBENCH_HUATUO_URL`
	`http://localhost:8013` / `:8014` / `:8015`. Optional services; not installed by the MVP.

```
THORAXBENCH_OLLAMA_URL=http://127.0.0.1:11434 \
  uv run thoraxbench serve --port 8000
```

Model setup

1. **Inspect the model YAML.**

   Find its adapter, tag, view and prompt under `configs/models/`. Example: `medgemma1.5-q4` expects Ollama tag `medgemma1.5:4b-it-q4_K_M`.

2. **Check model terms and hardware needs.**

   See `docs/model_cards.md` and `docs/model_selection.md`. A model ID is not a download guarantee. Gated terms remain separate from dataset licenses.

3. **Start the local model service.**

   If Ollama is installed but not running, start it using your local installation's service controls or `ollama serve`. If it is already running, use that instance.

4. **Check the installed tags.**

   ```
   ollama list
   uv run thoraxbench models
   ```

Configuration files, prompts, preprocessing and sampling affect results. Preserve version control and the recorded hashes. Remote/LAN inference is a design proposal only; do not bypass the loopback guard.

Stop safely. Keep the holdout sealed.

Before any real-study workflow

Verify the specific dataset's license status with `uv run thoraxbench datasets`. Acquisition requires an approved registry entry; never infer a license from the hosting site. Acquisition automation and the benchmark runner remain unfinished.

1. **Freeze the patient split before indexing or analysis.**

   ```
   uv run thoraxbench split lidc_idri
   ```

   Run this only for an approved, already downloaded cohort that has no split yet. Existing frozen splits refuse replacement. Do not regenerate them.

2. **Index only permitted development cases.**

   ```
   uv run thoraxbench index lidc_idri --limit 1
   ```

   Indexed development studies appear in the browser. The browser has no arbitrary-folder upload flow. The CLI can read a permitted study directory through its guards.

Never open `data/sealed/`, inspect holdout images, or copy sealed cases into development folders. No current MVP command is an approved holdout evaluation workflow. The future benchmark requires a frozen experiment and explicit unsealing.

Observed problem	Recovery action
Model unavailable	Check `ollama list`, the configured tag and loopback URL. Use `heuristic-hu` for the demo. Do not substitute a cloud endpoint.
Slow run / empty or invalid output	Preserve the error and result. Try a smaller synthetic vision request. Do not turn an abstention into a negative finding.
Study missing / guard rejection	Check the development split and index. A holdout rejection is intentional. Do not bypass it.
Disk full / cannot write	Check `df -h .` and `du -sh data`. Review disposable generated results/caches before removing anything. Preserve raw data, manifests and split files.
Old download link stops working	Download links are registered for the running server process. After restart, open the saved files under `data/results/ui/` directly or rerun the study.

Shutdown and handover

Press **Ctrl+C** in the server terminal. Completed result files remain on disk. Record the model, settings, version, result paths, warnings and next action before you hand work to another researcher. Never commit patient images, rendered CT slices, model weights or secrets.

Maintain this manual: edit `manual.html`, then regenerate the PDF from the repository root with `uv run python -m playwright pdf --paper-format A4 file://"$PWD"/manual.html manual.pdf`. Install Chromium with `uv run playwright install chromium` if needed. Check `docs/plan.md` for current scope; this manual contains no patient images.